

Jan 2021

Noone (2012)

equation (6) $\frac{l_D}{l} = \alpha q_D/q$

$$\Rightarrow l_D = \alpha l q_D/q$$

$$\text{RHS of (8)} = \frac{q_D + l_D}{q + \alpha l}$$

$$= \frac{q_D + \alpha l q_D/q}{q + \alpha l}$$

$$= q_D (1 + \alpha l/q) / (q + \alpha l)$$

$$= \frac{q_D}{q} \cancel{(q + \alpha l)} / \cancel{(q + \alpha l)}$$

$$= q_D/q$$

$$\therefore q_D/q = \cancel{q_D} \frac{q_D + l_D}{q + \alpha l}$$



his equation (8)